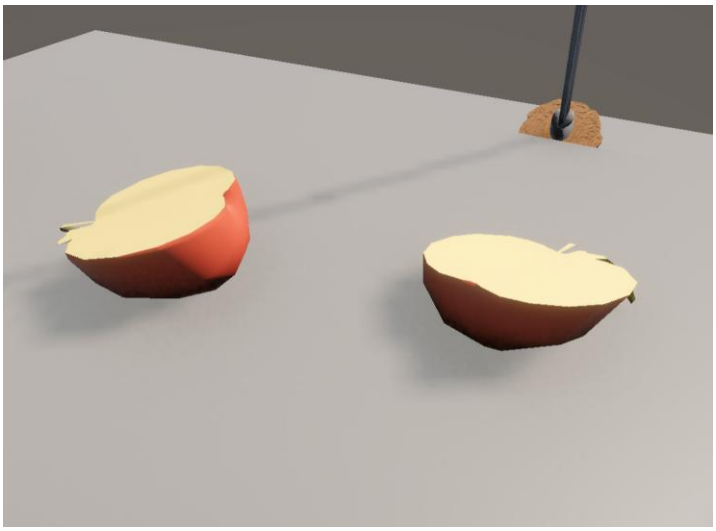
ELEC 328 Project Report
Assignment 1: Procedural Mesh Interactions

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1. Concept



This assignment was designed to give me experience dealing with vertex-level deformations on 3D meshes, including understanding the components of a mesh and knowing how to edit and generate a mesh procedurally. For this project, I created a system for cutting meshes reminiscent of Beat Saber and Fruit Ninja. My project contains three procedural mesh interactions. The first is the system to generate new meshes of an object's two halves when it is cut. The second is a continuous ripple effect to convey that a cut half is about to disappear. The third is a white trail that follows the player's sword to convey its speed and direction. Each of these deformations and interactions posed unique problems for me to solve, and I am very happy with how they came out.

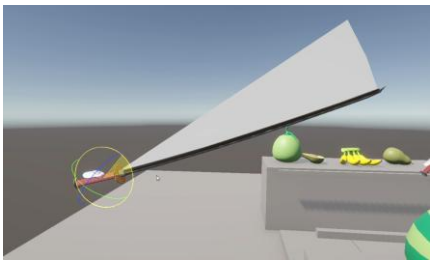
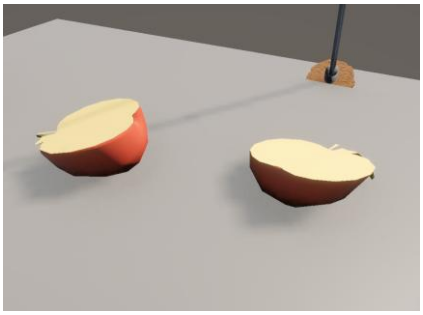
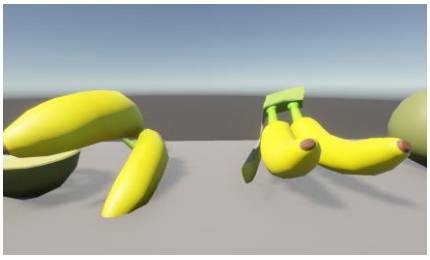
2. Technical overview

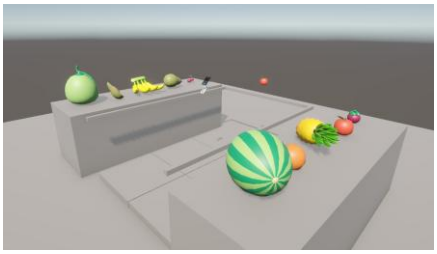
Explain the main system you built and the techniques that make it work. Use the prompts below that apply to your assignment.

The bulk of my work went towards making the mesh cutting system. This system takes a plane and finds which vertices of a mesh are above and below the plane. Vertices that are above the plane are grouped into one new mesh, and vertices below are grouped into a different mesh. Then, new vertices are generated at the intersection points of the plane and the mesh's triangles to create new triangles that make a smooth cut. Then, a new mesh is generated to cover the hole made by the cut, made by finding the center point between the new generated vertices and creating a cap using the center vertex and the edge vertices. I included a floating apple that when sliced resets all the fruits surrounding you so that you can continue to play with the cutting effect. The second system I made was a ripple system. This simply displaces a mesh's vertices along the y axis of the world, with a phase displacement based on the vertex's x position. The displacement increases over time, which gives the ripple

a mirage-like feeling. The final system I made is a sword trail system. The sword trail is a mesh with 6 vertices, and every frame the vertices are updated to follow behind the sword. This mesh doesn't move via transforms, instead it moves through moving its vertex positions.

3. Evidence of requirements

Requirement	Where it is demonstrated	Screenshot
Input-responsive deformation	Experience video 2:09-2:27 Narrated video 0:44-2:21 Narrated video 3:05-3:35	 The sword's trail responds to the movements of the player.  The player can use their movement input to cut the fruits.
Continuous deformation	Experience video 1:10-2:09 Narrated video 2:21-3:05	 After being cut, the fruit halves ripple before disappearing to give them an ethereal feeling when being de-spawned, instead of just disappearing immediately. Here the banana is mid ripple.

Spatial placement	Experience video 0:05-0:27	 Each of the fruits are placed on display outside the rails for the player to freely cut.

5. Assets, data, and citations

List every external asset, dataset, model, texture, sound, library, or reference used. State whether it was created by you, provided by the course, or obtained elsewhere. Provide a source link or formal citation where appropriate.

Item	Source / citation	How it was used
Reality Roost SDK	Course provided asset	The Reality Roost SDK is the foundation of the project, providing XR tracking that I used to let the player swing their swords.
Low Poly Fruit	https://assetstore.unity.com/packages/3d/vegetation/plants/free-pack-fruits-368828	I used the low poly fruit as the main interactable mesh.
Katanas	https://assetstore.unity.com/packages/3d/props/weapons/japanese-katana-287314	The katanas are the visual for how the player cuts the fruits.
Beat Saber Gameplay	https://www.youtube.com/watch?v=sjz1MU7mLhQ	I used this footage in my narrated video.
Fruit Ninja Gameplay	https://www.youtube.com/watch?v=DZa05028eJA	I used this footage in my narrated video.

6. AI-use disclosure

If you used generative AI during project development, disclose it below. If you did not use AI, write: No generative AI was used in project development.

Tool	Used for	What was incorporated, changed, or rejected
Claude	Debugging	I initially wrote the entire mesh cutting script without the use of AI, including deriving exactly what to do with vertices and how to create new ones. Then, once I had the meshes successfully being cut, albeit with bugs, I used Claude to clean up and debug various issues, including missing or misoriented triangles, incorrect texture mapping, and indexing errors. I also used it to help devise a solution to somewhat smooth the normals of the mesh after cutting. Originally, my solution to create the fruit halves had many duplicated vertices, which made the shading of the fruits not smooth. Claude was used to help develop a way to merge vertices that shared positions and UV coordinates such that they also shared normals. However, the fruits use a color tile texture to apply flat colors, so vertices on the edge of colored lines, like on the watermelon, could share positions but not UV coordinates, so the normal could not be easily shared.
Gemini	Debugging	I used Google Gemini to explain why the systems I wrote for the rippling fruit halves and sword trail didn't immediately work, which helped to clarify and solidify my understanding of the topics. For example, when making the fruit halves ripple, my original solution deformed the vertices in the local space of the fruit half, so any transformation I did caused the vertex to move way more than I intended. I had forgotten to transform the vertex positions to world space, which Gemini helped to point out.

7. Reflection

I am extremely happy with how well the mesh cutting system I made ended up working. The cutting is satisfying to do in VR and looks great visually. It does have some flaws, such as the sword's cut plane being based on its orientation and not its movement, making your cut angle not always match the direction you sliced. I also think that my mesh cutting script could use some work, as I continually added things to a single function instead of dividing things up. This made the file somewhat hard to navigate once it got long. One thing I attempted to explore but ultimately abandoned was making the fruit halves also cuttable. However, this proved difficult since I was creating two meshes (the outside and the inside of the fruit) per fruit half, so I couldn't simply drag-and-drop my existing script onto the halves. If I were to continue this project, that is where I would continue from. One thing I learned doing this project is how strict the mesh property arrays are, as simple off-by-one errors led to vastly incorrect results. I also learned new ways to handle getting rid of objects. Since the fruit halves were children of the original fruit, I couldn't simply delete the fruit. So, I made a script that hides the original fruit far away, which added the benefit of being able to easily bring it back when the player slices the reset apple.